
Problem Set 4

Econ 502: Advanced Microeconomics

Problem 1: Adverse Selection in the Used Laptop Market

A market for used laptops has two quality types:

- **High quality (H):** worth \$600 to sellers and \$1,000 to buyers
- **Low quality (L):** worth \$200 to sellers and \$400 to buyers

Let $\lambda \in [0, 1]$ denote the fraction of laptops that are high quality. Sellers know the quality of their own laptop; buyers cannot observe quality before purchase.

- a) **Symmetric information.** Suppose buyers can perfectly observe quality. Describe the competitive equilibrium. Which types trade?
- b) **Asymmetric information with $\lambda = 0.3$.** Buyers cannot observe quality and believe a fraction $\lambda = 0.3$ of laptops are high quality. What is a buyer's expected value of a laptop? Given this expected value, which sellers are willing to sell? What is the highest possible equilibrium price in this market?
- c) **Asymmetric information with $\lambda = 0.6$.** Repeat part (b) with $\lambda = 0.6$. Does the market unravel in this case? Explain why the outcome differs from part (b).

Problem 2: Moral Hazard in Unemployment Insurance

A worker is laid off and enters a two-period model. In period 1, she is unemployed and receives UI benefit b . She chooses search effort $e \geq 0$ at personal cost $c(e)$. With probability e , she finds a job paying wage w in period 2; with probability $1 - e$, she remains unemployed and receives b again in period 2. The worker has utility $u(c)$ with $u' > 0$ and $u'' < 0$ (strictly risk-averse). The cost of search function $c(e)$ is increasing and convex, with $c(0) = 0$, $c' > 0$, and $c'' > 0$.

- a) Write down the worker's expected utility as a function of e , b , and w . Derive the first-order condition for the optimal effort level e^* and give an economic interpretation.
- b) Show that $\frac{de^*}{db} < 0$. Explain intuitively why higher benefits reduce search effort.
- c) Now suppose $u(z) = \sqrt{z}$, $c(e) = \frac{1}{2}e^2$, and $w = 9$. Find $e^*(b)$ explicitly. At what benefit level does the worker exert zero effort?

Problem 3: Externalities and Regulatory Policy

A chemical plant produces q units of output per day and earns a profit:

$$\pi(q) = 80q - q^2$$

Its production generates pollution that imposes a total external cost on a neighboring farm of:

$$EC(q) = 10q$$

- a) Find the chemical plant's privately optimal output q_{priv} and the associated profit. What is the total external cost imposed on the farm at this output level?
- b) Find the socially efficient output q^* that maximizes total surplus (plant profit minus external cost). Calculate the deadweight loss from unregulated production.
- c) Derive the Pigouvian tax t^* per unit of output that induces the plant to choose the efficient output. Verify that the tax achieves the social optimum.
- d) **Coase Theorem.** Suppose bargaining between the plant and the farm is costless.
 - i. **Farm holds the right** to zero pollution. Explain how the two parties negotiate. What output level emerges? What is the range of feasible payments from plant to farm?
 - ii. **Plant holds the right** to pollute freely. Starting from the plant's privately optimal output, show that there are gains from trade from reducing output to q^* . Describe the negotiation and the range of feasible payments.
 - iii. Compare the outcomes in (i) and (ii). What does the Coase theorem predict, and how does the assignment of property rights matter?

Problem 4: Public Goods and Free Riding

Two roommates (1 and 2) share an apartment and jointly consume a “household quality” public good G (WiFi, cleaning supplies, shared subscriptions) funded by their individual contributions $g_i \geq 0$, so $G = g_1 + g_2$. Roommate 1 has income $Y_1 = 120$ and roommate 2 has income $Y_2 = 180$. Each roommate's budget constraint is $x_i + g_i = Y_i$, and their utility is:

$$U_i(x_i, G) = x_i \cdot G$$

where $x_i = Y_i - g_i$ is private consumption.

Part I: Decentralized Equilibrium

- a) Taking the other roommate's contribution as given, derive roommate i 's best response function $g_i^*(g_j)$. How does the best response depend on g_j ? What does this reveal about the free-rider problem?
- b) Find the Nash equilibrium contributions (g_1^*, g_2^*) , total public good G^* , private consumption x_i^* , and utility U_i^* for each roommate. Note any surprising feature of the equilibrium given the income difference.

Part II: Social Optimum

- c) A social planner maximizes total welfare $U_1 + U_2$ subject to the resource constraint $x_1 + x_2 + G = Y_1 + Y_2$. Find the efficient total public good G^{**} . (*Hint: Let $X = x_1 + x_2$ and note that $U_1 + U_2 = X \cdot G$.*)
- d) If the planner allocates private consumption equally ($x_1^{**} = x_2^{**}$), find each roommate's contribution g_i^{**} and utility U_i^{**} . Compare G^* to G^{**} : by what fraction does decentralized provision fall short of the efficient level? Is each roommate better or worse off at the social optimum?

Part III: Policy

- e) Suppose the government subsidizes contributions at rate s : each dollar contributed costs the roommate only $(1 - s)$ dollars (the government covers the rest). Find the subsidy rate s^* that induces the efficient total public good G^{**} in the Nash equilibrium.

Problem 5: Game Theory

Part I: Normal Form Games

Consider the following simultaneous-move game. The row player is A and the column player is B.

	L	R
U	(6, 2)	(0, 4)
D	(3, 3)	(5, 1)

- a) Identify any strictly dominant or strictly dominated strategies. Eliminate any dominated strategies and explain your reasoning.
- b) Find all pure-strategy Nash equilibria, or explain why none exist.
- c) Find the unique mixed-strategy Nash equilibrium. Let p = probability A plays U and q = probability B plays L. Derive p^* and q^* and compute the expected payoff for each player.

Part II: Sequential Games

- d) Now modify the game to be sequential: Player A moves first (choosing U or D), then Player B observes A's choice and responds. Draw the game tree. Using backward induction, find the subgame perfect equilibrium (SPE). What is the equilibrium outcome and what are the payoffs?
- e) Compare the SPE outcome to the mixed-strategy equilibrium payoffs from part (c). Is moving first an advantage or a disadvantage for Player A in this game?